

Regularization and Stability

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1 Introduction

In the field of machine learning, understanding the generalization capabilities of learning algorithms is a fundamental challenge. Bousquet and Elisseeff (2002) proposed a framework based on the concept of *Algorithmic Stability*. This framework suggests that an algorithm will generalize well if its predictions remain uniformly stable when a single training example is removed or altered.

The main objective of this project is to empirically investigate the effect of the regularization strength on the algorithmic stability. Furthermore, generalization performance of different models is examined. This study focuses on Ridge Regression and unregularized least squares.

To ensure a robust empirical analysis, the stability metrics were evaluated on two distinct datasets: the real-world California Housing dataset, which presents natural collinearity and noise, and a Synthetic Dataset using samples from gaussian distribution for features, weights, and noises.

2 Datasets

2.1 Real-world Regression Dataset: California Housing

This data is sourced from sklearn.datasets library and contains geographical and demographic features used to predict the median house value for various California districts. The raw dataset consists of 20640 observations and 8 features. Features are as following:

1. **MedInc:** The median income of households within the block group, typically measured in tens of thousands of US dollars.
2. **HouseAge:** The median age of the houses within the block group, measured in years.
3. **AveRooms:** The average total number of rooms per household.
4. **AveBedrms:** The average number of bedrooms per household.
5. **Population:** The total number of residents living within the block group.
6. **AveOccup:** The average number of individuals residing in a single household.
7. **Latitude:** The geographical coordinate specifying the north-south position of the block group.
8. **Longitude:** The geographical coordinate specifying the east-west position of the block group.

The target value in this dataset is the median house value for California districts, measured in hundreds of thousands of dollars.

2.2 Synthetic Dataset

To further investigate the algorithmic stability, a synthetic dataset was employed in which the target value was generated through the following process:

$$Y = Xw + b + \epsilon$$

The components were engineered utilizing the following considerations:

1. **Independent Feature Space (X):** The predictor variables were sampled from an independent normal distribution ($\mathcal{N}(\mu, \sigma_x^2)$). Consequently, the absolute absence of correlated

features allows us to test the behavior of the L_2 penalty. Using this dataset, the ability of Ridge regression is examined on a fundamentally clean data.

2. True Weights (w), Induced Sparsity, and Bias: The variance of the weights was considered independently of the feature variance. It was sampled from $\mathcal{N}(\mu, \sigma_w^2)$. After constructing the weights vector, a portion of them (0.1) were randomly selected to be set to zero. This approach was employed to inject some induced sparsity which mimics some real world scenarios, for example there could be some features in a dataset which are not really relevant to the target value but are passed to the algorithm. To shift the values of target variable from origin, a static bias term was also added.

3. Dynamic Signal-to-Noise Ratio (SNR): Without any noise, the models easily capture the true weights. To avoid this, some noise (ϵ) should be injected to the target value. To generate normal error terms, an appropriate value for σ_{noise}^2 was needed. To choose this value dynamically, based on the values of the deterministic part of the target value, a measure called SNR was used. The process was as follows:

$$\sigma_{noise} = \sqrt{\frac{Var(Xw + b)}{\text{Target SNR}}}$$

By fixing the SNR at a targeted threshold, the random noise would be strong enough with respect to the deterministic values of Y .

To have a large number of observations comparable to the real-world dataset, 20,000 rows with 10 different features were produced. For features $N(0, 0.5^2)$, and weights $N(0, 1^2)$ were used. Also, bias term and Target SNR were set equal to 2.

3 Exploratory Data Analysis

In this section, EDA was conducted on both Real-world and Synthetic Datasets to explore characteristics of features and target values. This then ensures that proper pre-processing techniques would be applied in order to prepare the data to be suitable for the learning algorithms.

3.1 Real-world Dataset

3.1.1 Target Value Histogram

By plotting the histogram of the target value, the distribution of this variable can be seen. By doing this in Figure 1a, it was seen that there was a spike in the last bin, which corresponds to values around 5 (\$500,000).

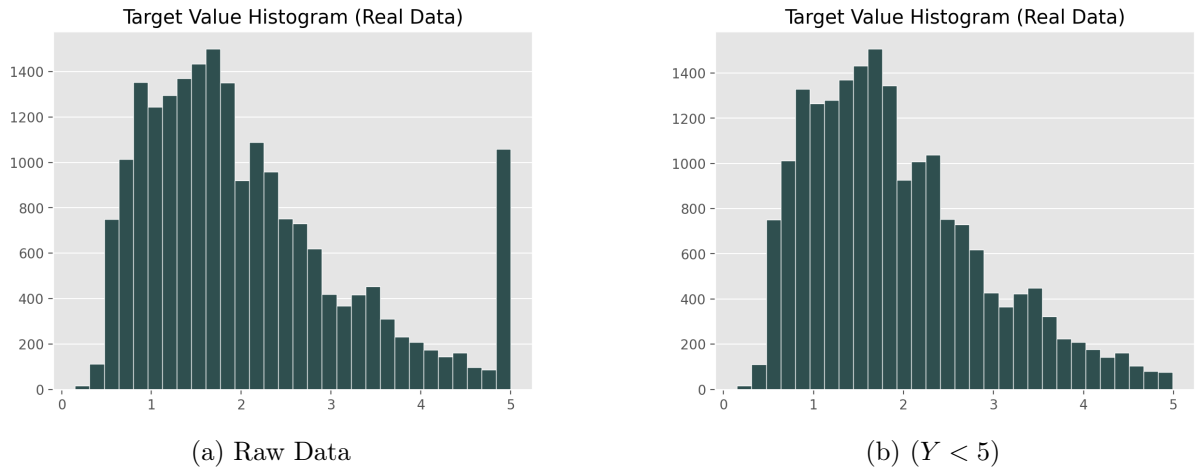


Figure 1: Histogram of Target Value in Real-World Dataset

The reason is that, for the purpose of this dataset, values above 5 were capped at 5, causing the violation of the assumption of continuous linear relationship between the features and the target. To ensure these capped values do not inflate the prediction errors, observations with $Y \geq 5$ were filtered out. The histogram for the filtered dataset is shown in Figure 1b. During this process, 992 observations were omitted.

3.1.2 Features Correlation Matrix

To better understand the structure of California Housing dataset, a Correlation Matrix Analysis was conducted. As Figure 2 reveals, there are some high multicollinearity between some features. The most negative correlation belongs to the pair of spatial variables, which with respect to the structure of California was expected. On the other hand, AveRooms and AveBedrms variables demonstrate the most positive correlation, which again was anticipated. Such multicollinearities can reveal the advantage of Ridge compared to unregularized model for handling such anomalies in the dataset.

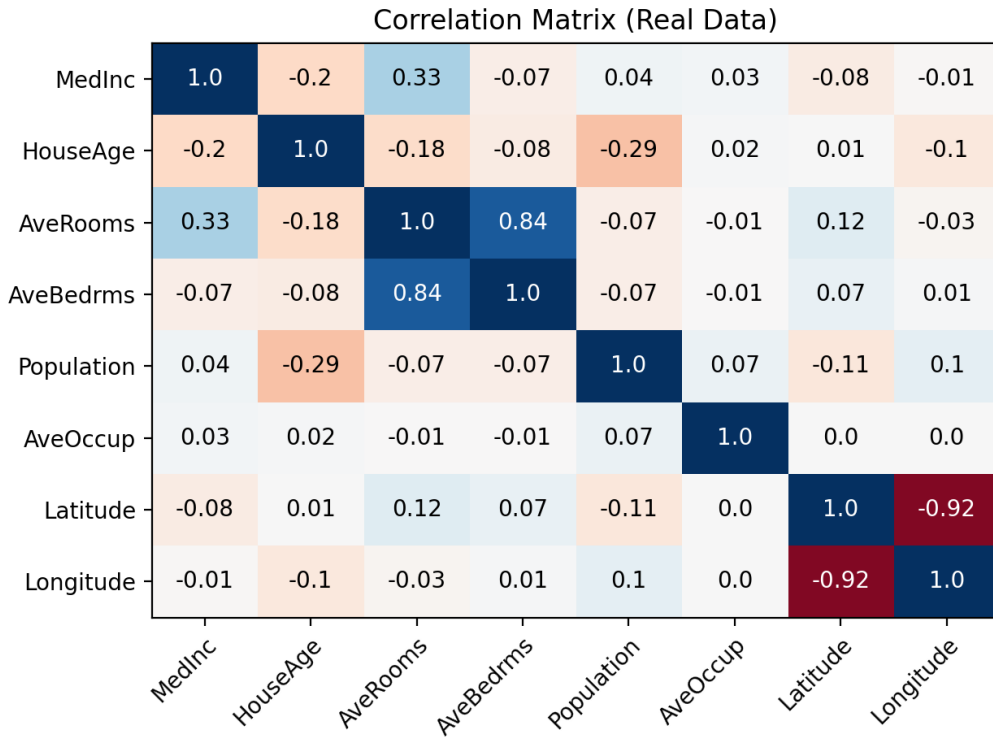


Figure 2: Correlation Matrix of Features of California Housing Dataset

3.1.3 Features Distribution

To ensure the need for some pre-processing techniques, such as standardization of features, the box plot of each feature is illustrated in Figure 3. This analysis confirms the different scales of different variables which is bad for training the model. As the Ridge regression applies a uniform L_2 penalty, applying the algorithm on raw and unscaled dataset will result in bias. For instance, Population variable ranges from 0 to near 35000 while MedInc variable ranges between 0 to 15 approximately.

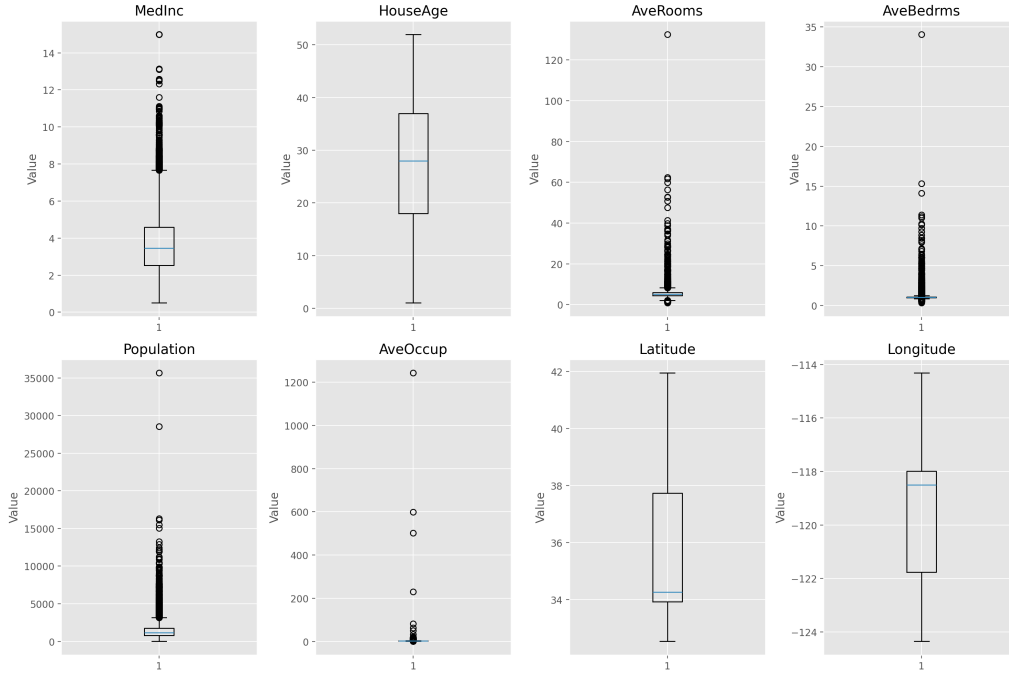


Figure 3: Box Plot for Each Feature of California Housing Dataset

3.2 Synthetic Dataset

3.2.1 Target Value and Noise Histogram

Figure 4 illustrates the distributions of the generated synthetic target variable and the injected Gaussian noise. As shown in panel (4b), the additive noise follows a normal distribution centered at zero ($\mu = 0$), providing the essential random variance required to evaluate model overfitting. Panel (4a) confirms that by deliberately calibrating the standard deviations of the features and weights, alongside a predefined intercept, the final synthetic target values were successfully constrained to a realistic numerical scale. This intentional design choice ensures that the synthetic experiment remains numerically stable and directly comparable to the empirical bounds of the real-world California Housing dataset.

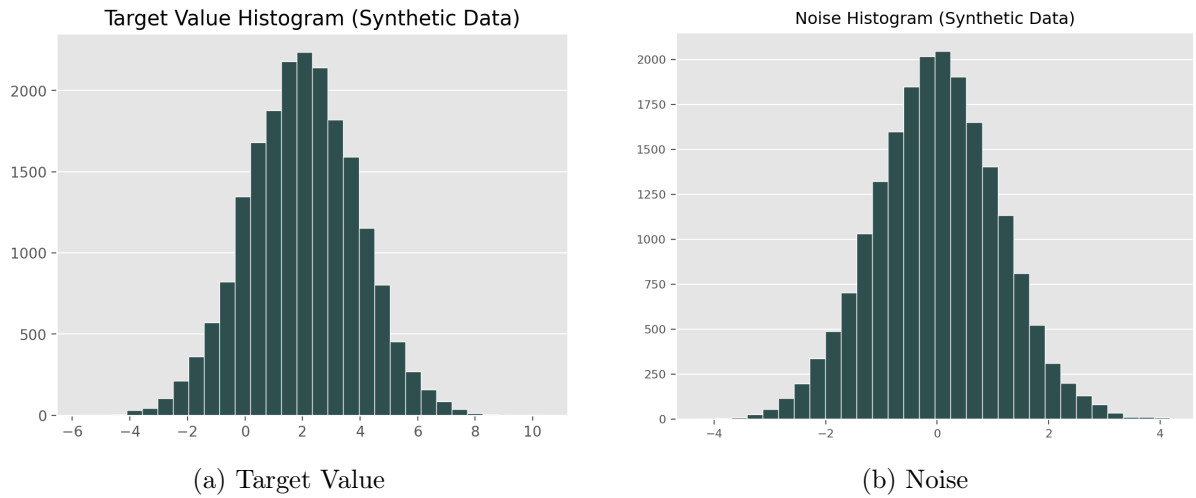


Figure 4: Histogram of Target Value and Noise in Synthetic Dataset

3.2.2 Features Correlation Matrix

To explore the relationships among the generated variables, a Pearson correlation matrix was constructed. By plotting this matrix in Figure 5, it was seen that the off-diagonal elements were uniformly near zero. This confirms that the Data Generating Process (DGP) successfully produced independent, uncorrelated features. Without the severe multicollinearity observed in the real-world dataset, the unregularized design matrix is inherently well-conditioned.

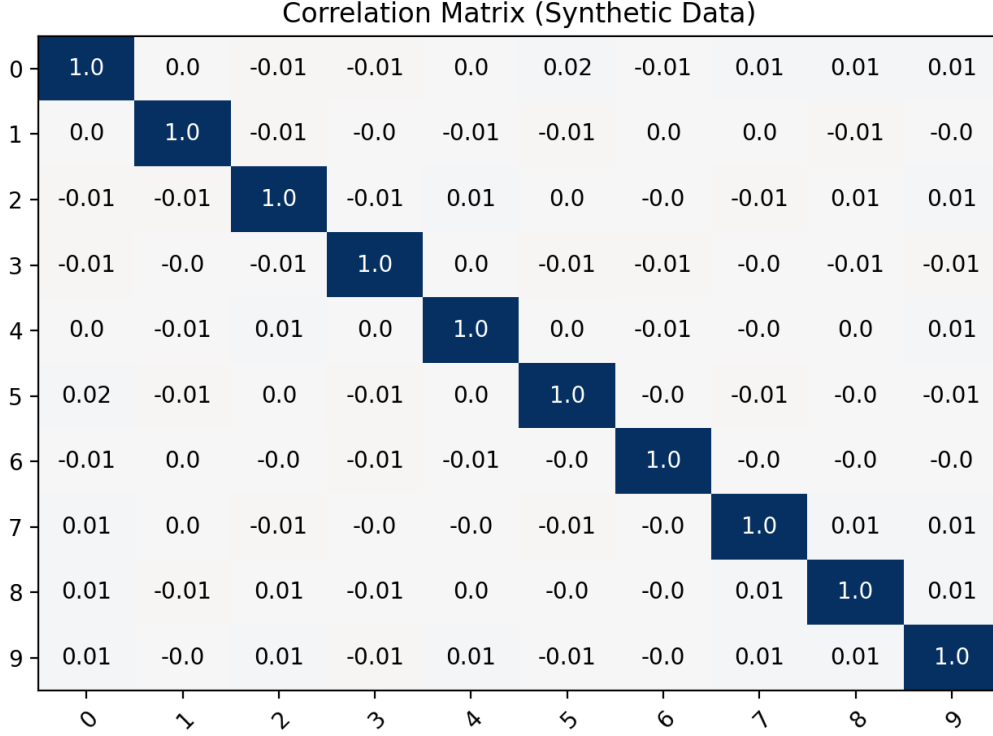


Figure 5: Correlation Matrix of Features of Synthetic Dataset

4 Pre-Processing Techniques

To ensure mathematical integrity and algorithmic stability, the dataset required strict preprocessing before model evaluation. First, instances where the target variable was artificially capped at its maximum value were filtered out to remove non-linear anomalies and preserve the underlying continuous distribution. Second, because the Ridge penalty applies uniformly to all learned weights, the feature space was standardized to a zero mean and unit variance to prevent features with naturally larger numerical ranges from disproportionately skewing the objective function. Crucially, to strictly prevent data leakage, these standardization parameters were derived exclusively from the training subsamples and subsequently applied to the universal test set.

It is worth mentioning that standard linear regression models are unconstrained and can theoretically output negative predictions for strictly positive real-world variables, such as housing prices. While applying a logarithmic transformation to the target variable ($Y' = \ln(Y)$) is a standard way to prevent this, the raw target scale was intentionally maintained for this experiment. This choice ensures that the Test MSE and the Algorithmic Stability metric (Average Prediction Change) remain easily interpretable and linearly comparable across all regularization strengths (λ).

4.1 Data Splitting and the Universal Test Set

To ensure a mathematically rigorous evaluation of generalization error and strictly prevent data leakage, a universal hold-out test set was established prior to any model training or feature standardization. The full dataset was partitioned, reserving 20% exclusively for testing, while the remaining 80% formed the sampling pool for the training subsets. By fixing the random seed during this initial split, it was guaranteed that all subsequent models—regardless of their training subsample size (N) or regularization strength (λ)—were evaluated against the exact same unseen data. This structural choice ensures that observed variations in the Test Mean Squared Error are entirely attributable to the algorithmic stability and complexity of the model, rather than arbitrary differences in the testing distribution. Furthermore, to maintain the integrity of the test set, standardization statistics (mean and variance) were computed strictly on the active training subsample and subsequently applied to scale the universal test set.

5 Implementation and Methodology

5.1 The Core Algorithm: Ridge Regression

To mitigate the high variance associated with multicollinearity, Ridge regression (L_2 regularization) was implemented as the primary learning algorithm. Unlike standard ordinary least squares, Ridge regression applies a penalty to the squared magnitude of the coefficients. To compute the optimal weights (w), the objective function was defined to minimize the residual sum of squares subject to the regularization strength (λ). The closed-form solution was computed using the following normal equation:

$$w = (X^T X + \lambda I)^{-1} X^T Y \quad (1)$$

To ensure the penalty was applied symmetrically, the design matrix X was standardized, and the bias term was explicitly excluded from the regularization identity matrix (I).

The linear system was solved using `numpy.linalg.solve` rather than explicitly computing the matrix inverse for improved numerical stability.

5.2 Leave-One-Out Stability Analysis

To empirically evaluate algorithmic stability, a Leave-One-Out Cross-Validation (LOOCV) pipeline was developed. For a given training subset of size N , the base model was initially fit to the entire subset to establish the full-data weights. Subsequently, an iterative process was executed where exactly one observation (i) was removed, and the model was refit to the remaining $N - 1$ samples. To optimize computational efficiency during this iterative process, the matrices $X^T X$ and $X^T Y$ were precomputed on the full dataset, allowing the LOO weights to be derived via rapid matrix subtraction rather than complete recalculation.

5.3 Evaluation Metrics

To quantify model behavior across varying regularization strengths and dataset sizes, three primary metrics were calculated:

- **Empirical Error (Train MSE):** The mean squared error computed on the specific training subsample used to fit the model.
- **Generalization Error (Test MSE):** To prevent data leakage and provide a mathematically fair comparison across all models, the test error was evaluated exclusively on a universal, hold-out test set (20% of the original data) that remained unseen during the training phase.

- **Algorithmic Stability:** Following the theoretical framework of Bousquet and Elisseeff, stability was measured by calculating the average absolute difference in predictions when a single data point was removed from the training set. For a universal test set of size M , the stability metric was formalized as:

$$\text{Stability} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M |\hat{y}_j - \hat{y}_j^{(-i)}| \right) \quad (2)$$

where $\hat{y}_j$ is the prediction of the full model, and $\hat{y}_j^{(-i)}$ is the prediction of the model trained without the i -th instance.

6 Empirical Results and Visualizations

To empirically validate the theoretical bounds proposed by Bousquet and Elisseeff, the behavior of the Ridge regression model was evaluated across a range of training subsample sizes (N) and regularization strengths (λ). The results were compared between the empirical California Housing dataset and the strictly controlled synthetic dataset.

6.1 Regularization and the Bias-Variance Tradeoff

To observe the impact of L_2 regularization on model generalization, the Test Mean Squared Error (MSE) was plotted against λ for both datasets (Figure 6). For the real-world dataset, it was observed that models trained on small subsamples ($N \in \{100, 500, 1000\}$) exhibited severe overfitting, resulting in massive initial test errors. However, as the regularization penalty was increased, the Test MSE for these low-data regimes decreased. This confirms that regularization successfully restricts complex weights and cures high variance.

Conversely, for the synthetic dataset, a distinct pattern emerged. Because the Data Generating Process produced inherently independent features without the multicollinearity present in the real-world data, the unregularized models were already stable. Consequently, it was observed that applying aggressive regularization to the synthetic data crushed the true weights toward zero. As $\lambda \rightarrow 10^4$, all models, regardless of dataset size, converged toward a high-bias underfitting state, successfully illustrating the classic Bias-Variance tradeoff.

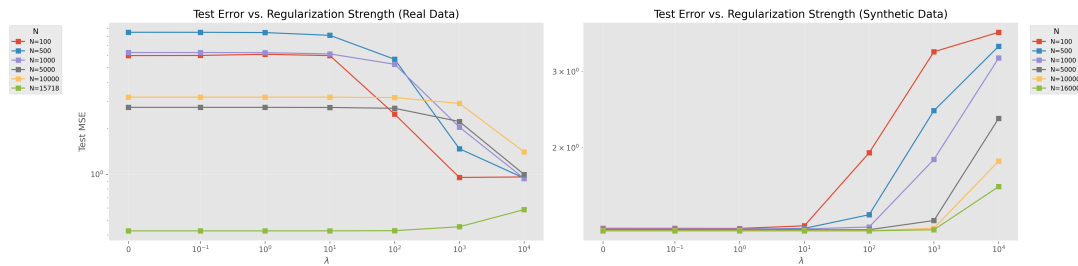


Figure 6: Test Error vs. Regularization Strength for Real and Synthetic Datasets.

6.2 Algorithmic Stability vs. Regularization Strength

To quantify algorithmic stability, the average prediction change under Leave-One-Out Cross-Validation (LOOCV) was plotted against λ (Figure 7). For both datasets, it was seen that the application of Ridge regression led to a decrease in average prediction change. Mathematically, for large values of λ , the weights are heavily penalized toward zero, so the model relies on the unregularized baseline mean ($\bar{Y}$). Because the sample mean changes minimally when one observation is removed, extreme regularization makes the model highly stable.

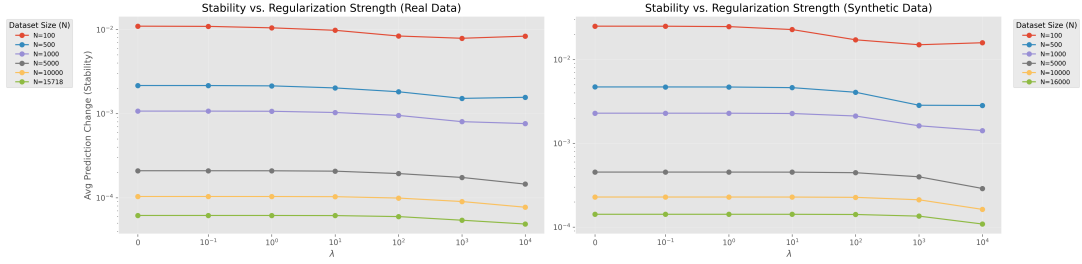


Figure 7: Average Prediction Change (Stability) vs. Regularization Strength.

6.3 Generalization Gap vs. Stability

To connect algorithmic stability to generalization, the generalization gap (defined as the absolute difference between Test MSE and Train MSE) was evaluated (Figure 8). For both datasets, it was seen that models with a large gap in the unregularized state converged toward zero as λ increased. By restricting the weights, the L_2 penalty forces the model to generalize uniformly. This shows that algorithmic stability bounds the generalization gap.

However, an exception was seen in the synthetic dataset for small subsamples (e.g., $N = 500$), where the gap increased at extreme values of λ . Because extreme regularization forces the model to continuously predict the training mean, the Test MSE became much larger than the Train MSE. This occurred because the baseline mean of a small training subsample does not perfectly match the true mean of the universal test set, causing the gap to widen due to severe underfitting.

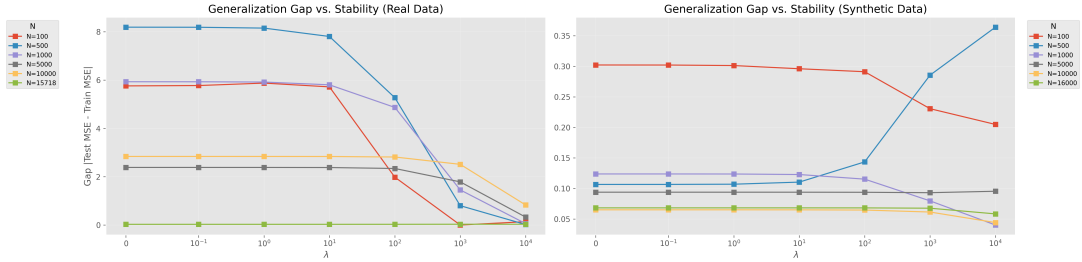


Figure 8: Generalization Gap vs. Regularization Strength (λ).

6.4 Effect of Dataset Size on Stability

Finally, to explore the relationship between dataset size and stability, the average prediction change was plotted against N on a logarithmic scale for different values of λ (Figure 9). A linear downward trend was seen for all configurations in both experiments. This empirically validates the theoretical bound of $O(1/N)$. It is concluded that having a large number of observations acts as a natural regularizer, reducing variance without needing an artificial penalty term.

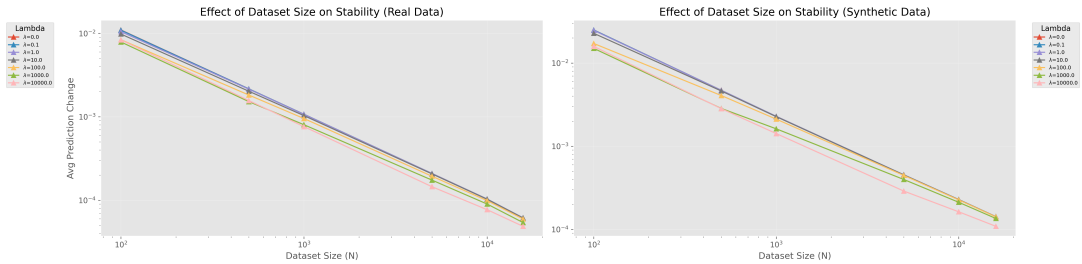


Figure 9: Effect of Dataset Size (N) on Stability, plotted on a log-log scale.

7 Conclusion

In this study, the theoretical framework regarding algorithmic stability and generalization error was empirically validated. By evaluating the real-world dataset, it was seen that the improvement in generalization depends heavily on the sample size (N). For small sample sizes ($N \leq 500$), unregularized models showed severe overfitting because of highly correlated features. In these cases, applying Ridge regression yielded a significant improvement, making the models stable and reducing the generalization gap to zero.

Conversely, when a large number of observations was used ($N \approx 16,000$), the variance of the unregularized models was naturally low. Based on the $O(1/N)$ bound, the large dataset acted as a natural regularizer. Therefore, adding an L_2 penalty in this scenario provided no significant predictive improvement and caused underfitting when extreme penalties were applied. Ultimately, it was concluded that while algorithmic stability is required for generalization, explicit regularization is only practically significant when the dataset is small.